
PropCRM

Lead Management System for Real Estate Agents

USER FLOWS & SYSTEM TRIGGERS

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1. Document Purpose

This document defines the complete user flows for both actors (Customer and Agent) within the PropCRM system, along with all automated system triggers and their conditions. It serves as the definitive reference for understanding **who does what, when, and what happens as a result**.

1.1 Actors

ACTOR	TYPE	DESCRIPTION
CUSTOMER	Human — External	A property seeker who submits an inquiry through the lead form
AGENT	Human — Internal	The real estate agent who manages leads through the dashboard
SYSTEM	Automated	PropCRM backend (Google Apps Script) — executes business logic, sends emails
SCHEDULER	Automated	Time-driven trigger in Apps Script — runs periodic tasks

2. Customer User Flows

Flow 2.1: Submit Property Inquiry

Actor: **CUSTOMER**

Entry Point: Lead Form URL

Est. Duration: 30–60 seconds

1 Customer Opens Lead Form

Customer navigates to the PropCRM lead form URL (hosted on GitHub Pages). The page loads with a clean, glassmorphism-styled form on a cream-green background.

2 Customer Fills Form Fields

Customer enters their

Full Name

,

Email Address

,

Property Area

(locality — free text), and

Property Price

(budget in ₹). Each field validates in real-time as the user types or moves focus.

⚡ CONDITION: CLIENT-SIDE VALIDATION

- **IF** Name is empty or < 2 characters → Show error: "Please enter your full name (min 2 characters)"
- **IF** Email format is invalid → Show error: "Please enter a valid email address"
- **IF** Area is empty → Show error: "Please enter the property area/locality"
- **IF** Price is empty, zero, or negative → Show error: "Please enter a valid property price"
- **IF** any validation fails → Submit button remains disabled / submission is blocked

3 Customer Clicks "Submit Inquiry"

Submit button changes to a loading state (spinner animation). The button is disabled to prevent double submission. Form data is serialized as JSON.

4 [SYSTEM] Processes Submission

Data is sent to Google Apps Script via HTTPS POST. Server-side validation runs. If valid: lead is saved to Google Sheets, acknowledgment email is sent to the customer, notification email is sent to the agent. *(See Data Flow Document — Flow 1 for full details)*

⚡ CONDITION: SUBMISSION RESULT

- IF success → Go to Step 5a (Success)
- IF server validation error → Go to Step 5b (Validation Error)
- IF network/server error → Go to Step 5c (System Error)

5a Success: Confirmation Displayed

A green success toast appears: *"Thank you! Your inquiry has been submitted. Our agent will contact you shortly."* Form fields are cleared. Customer can submit another inquiry.

5b Validation Error: Errors Displayed

An error toast appears with the server-returned validation errors. Form data is preserved — the customer corrects the errors and re-submits.

5c System Error: Retry Prompt

An error toast appears: *"Something went wrong. Please try again."* Form data is preserved. Submit button is re-enabled for retry.

6 Customer Receives Acknowledgment Email

Within 60 seconds, the customer receives an email from PropCRM confirming their inquiry was received, with a summary of their submitted details. *(Sent via Brevo API from Apps Script)*

Flow 2.2: Customer Receives Follow-up Email

Actor: **CUSTOMER** (passive)

Trigger: Agent sends email from dashboard

1 [AGENT] Clicks "Send Email" on Dashboard

Agent identifies a lead in the dashboard table and clicks the "Send Email" action button.

2 [SYSTEM] Sends Follow-up Email via Brevo

Apps Script constructs a follow-up email using a predefined template, personalized with the lead's name, area, and price. The email is sent via Brevo API to the lead's email address.

3 Customer Receives Follow-up Email

The customer receives a professional follow-up email from the agent regarding their property inquiry. The email includes the agent's name and contact details.

3. Agent User Flows

Flow 3.1: Dashboard Login

Actor: **AGENT**

Entry Point: Dashboard URL

Est. Duration: 5–10 seconds

1 Agent Opens Dashboard URL

Agent navigates to the dashboard page. A PIN login overlay/modal is displayed covering all dashboard content.

2 Agent Enters PIN

Agent types their numeric PIN in the input field and clicks "Login" (or presses Enter).

3 [SYSTEM] Verifies PIN

PIN is sent to Apps Script via POST. Apps Script compares it against the PIN stored in AgentConfig sheet.

⚡ CONDITION: PIN VERIFICATION RESULT

- **IF** PIN matches → Login overlay is removed. Dashboard loads with full data. Go to Flow 3.2.
- **IF** PIN does not match → Error message: "Invalid PIN. Please try again." PIN input is cleared. Agent retries.

4 Dashboard Loads

Upon successful login, the system fetches all leads data and renders the complete dashboard: stats cards, charts, and leads table. The PIN is stored in a JavaScript variable (session memory) for subsequent API calls.

Flow 3.2: View Leads & Analytics

Actor: AGENT

Prerequisite: Successful login (Flow 3.1)

1 [SYSTEM] Fetches & Renders Data

GET request to Apps Script with PIN. System returns all leads + aggregated stats. Dashboard renders:

COMPONENT	DATA DISPLAYED	VISUAL STYLE
Stat Card 1	Total Leads (count)	Large number with icon
Stat Card 2	Contacted % (percentage)	Circular progress or percentage
Stat Card 3	Uncontacted (count)	Highlighted if > 0 (attention needed)
Stat Card 4	Average Property Price (₹)	Formatted currency
Chart 1	Price Bracket Distribution	Doughnut chart with green gradient + % labels
Chart 2	Area/Locality Distribution	Horizontal bar chart with green gradient
Leads Table	All leads with details	Sortable table with action buttons

2 Agent Reviews Dashboard

Agent scans stat cards for quick summary, reviews charts for market insight, and scrolls through the leads table. Agent can click "Refresh" at any time to fetch the latest data.

Flow 3.3: Mark Lead as Contacted

Actor: **AGENT**

Prerequisite: Dashboard loaded

1 Agent Checks the "Contacted" Checkbox

In the leads table, the agent clicks the checkbox next to a lead they have contacted. The checkbox immediately shows a loading state.

2 [SYSTEM] Updates Lead Status

POST request sent to Apps Script with action: "updateContacted", the lead ID, and contacted: true. Apps Script finds the row in Google Sheets and sets contacted = TRUE, contactedAt = current timestamp.

⚡ CONDITION: UPDATE RESULT

- **IF success** → Checkbox shows checked state. Row may visually update (subtle green tint). Stats cards re-calculate (Contacted % increases, Uncontacted count decreases).
- **IF failure** → Checkbox reverts to unchecked. Error toast shown: "Failed to update. Please try again."

3 Agent Can Uncheck (Reverse Contact Status)

If the agent unchecks the checkbox, the same flow runs in reverse — contacted is set back to FALSE and contactedAt is cleared.

Flow 3.4: Send Follow-up Email to Lead

Actor: **AGENT**

Prerequisite: Dashboard loaded

1 Agent Clicks "Send Email" Button

In the leads table, the agent clicks the "Send Email" action button on the row of the lead they wish to follow up with. The button changes to a loading spinner.

2 [SYSTEM] Sends Follow-up Email

POST request sent with the lead ID. Apps Script retrieves the lead's data and the agent's name from config. A predefined follow-up email template is populated and sent to the lead's email via Brevo API.

⚡ **CONDITION: EMAIL SEND RESULT**

- **IF** email sent successfully → Button shows ✓ "Sent" state for 3 seconds, then reverts to normal.
- **IF** email fails → Button shows X "Failed" state. Error toast: "Email could not be sent. Please check your Brevo configuration."

Flow 3.5: Agent Receives New Lead Notification

Actor: **AGENT**

(passive)

Trigger: New lead submitted

1 [CUSTOMER] Submits Lead Form

A customer fills out and submits the lead form.

2 [SYSTEM] Sends Notification Email to Agent

As part of the lead submission flow, Apps Script sends an email to the agent's email (from AgentConfig) with the lead's details: name, email, area, price, and a link to the dashboard.

3 Agent Reviews Email Notification

Agent receives the email in their inbox. They can see the lead's details and choose to either contact the lead directly or open the dashboard for full management.

4. Automated System Triggers

These are actions performed by the system **without any human intervention**. They are triggered by events or scheduled timers.

Trigger 4.1: New Lead Submitted (Event-Driven)

Type: **EVENT TRIGGER**

Fires: Immediately on form submit

PROPERTY	DETAIL
Event	Customer submits the lead form (POST submitLead)
Condition	Server-side validation passes (all fields valid)
Actions Executed	1. Generate UUID and timestamp 2. Calculate price bracket from price value 3. Write lead record to "Leads" sheet 4. Send acknowledgment email to customer (via Brevo) 5. Send new lead notification email to agent (via Brevo) 6. Return success response to frontend
Failure Handling	If email sending fails, the lead is still saved. Email flags remain FALSE — no data is lost.
Performance Target	Complete execution in < 3 seconds

Trigger 4.2: Uncontacted Lead Reminder (Time-Driven)

Type:

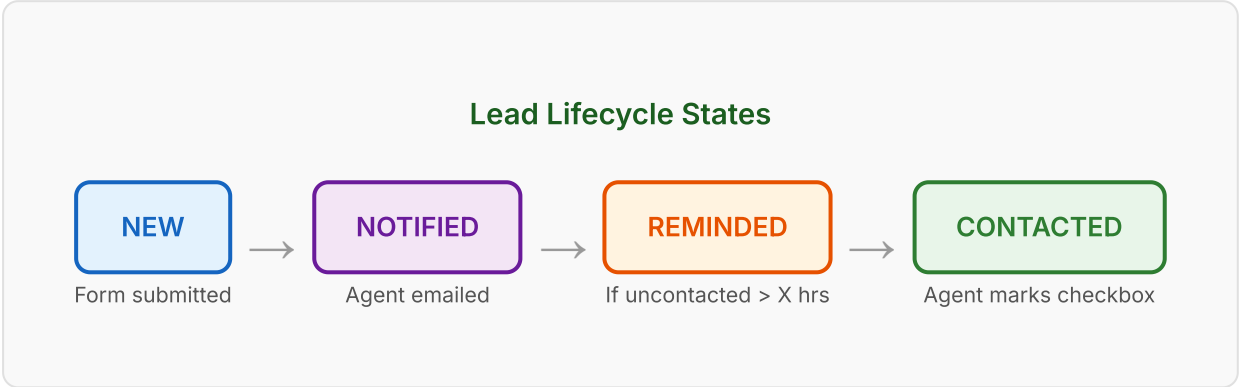
SCHEDULED TRIGGER

Fires: Every 1 hour
(configurable)

PROPERTY	DETAIL
Schedule	Google Apps Script time-driven trigger, executes every 1 hour
Function	checkUncontactedLeads()
Query Conditions	Find all leads in "Leads" sheet WHERE: contacted = FALSE AND reminderSentToAgent = FALSE AND submittedAt < (NOW - reminderHours)
For Each Matching Lead	1. Construct reminder email with lead details and urgency message 2. Send reminder email to agent's email via Brevo 3. Set reminderSentToAgent = TRUE for this lead
Idempotency	Reminder is sent only once per lead — the reminderSentToAgent flag prevents duplicate emails
Configuration	reminderHours is read from AgentConfig sheet (default: 1 hour)

5. Lead State Machine

Every lead progresses through a defined set of states. Transitions between states are triggered by system events or agent actions.



5.1 State Transition Table

CURRENT STATE	TRIGGER EVENT	CONDITION	NEXT STATE	ACTIONS
—	Customer submits form	Validation passes	NEW	Save to Sheets, send ack email to customer, send notification to agent
NEW	Notification email sent	Brevo API succeeds	NOTIFIED	Agent has been informed via email
NOTIFIED	Reminder trigger fires	contacted = FALSE AND reminderSent = FALSE AND age > reminderHours	REMINDED	Send reminder email to agent, set reminderSent = TRUE
NEW / NOTIFIED / REMINDED	Agent checks	Agent is authenticated (PIN)	CONTACTED	Set contacted = TRUE, record

CURRENT STATE	TRIGGER EVENT	CONDITION	NEXT STATE	ACTIONS
	"Contacted" checkbox			contactedAt timestamp
CONTACTED	Agent unchecks "Contacted" checkbox	Agent is authenticated (PIN)	NOTIFIED or REMINDED	Set contacted = FALSE, clear contactedAt. Note: reminder flag is NOT reset.

Note: The states NEW, NOTIFIED, REMINDED are implicit — derived from the combination of boolean flags (contacted, reminderSentToAgent) and timestamps. There is no separate "state" column in the sheet; the state is computed logically.

6. Complete Event-Trigger Matrix

This matrix provides a comprehensive, single-reference view of **every event in the system**, who triggers it, what conditions apply, and what actions result.

#	EVENT	TRIGGERED BY	CONDITIONS	SYSTEM ACTIONS	EMAILS SENT
E1	Form page loaded	CUSTOMER	None	Render form UI	None
E2	Form field validated	CUSTOMER	Field loses focus or value changes	Run client-side validation, show/hide error messages	None
E3	Form submitted	CUSTOMER	All client-side validation passes	POST to Apps Script → Server validates → Save to Sheets → Return response	1) Ack to customer 2) Notification to agent
E4	Dashboard opened	AGENT	None	Show PIN login overlay	None
E5	PIN submitted	AGENT	PIN matches AgentConfig	Hide login overlay → Fetch all leads → Render dashboard	None
E6	Refresh clicked	AGENT	Agent is authenticated	Re-fetch leads data → Re-render	None

#	EVENT	TRIGGERED BY	CONDITIONS	SYSTEM ACTIONS	EMAILS SENT
				stats, charts, table	
E7	Contacted checkbox toggled	AGENT	Agent is authenticated	POST update → Modify Sheet row → Update UI	None
E8	Send Email clicked	AGENT	Agent is authenticated	POST request → Fetch lead data → Construct email → Send via Brevo	1) Follow-up to customer
E9	Hourly timer fires	SCHEDULER	Leads exist where: contacted = FALSE AND reminderSent = FALSE AND age > X hrs	For each qualifying lead: send reminder → set flag	1) Reminder to agent (per qualifying lead)

7. Edge Cases & Error Scenarios

#	SCENARIO	EXPECTED BEHAVIOR
EC1	Customer submits form while offline / network drops mid-request	Fetch fails → Error toast shown → Form data preserved → User retries when online
EC2	Customer double-clicks submit button	Button is disabled on first click → Only one request sent
EC3	Customer submits same email multiple times	v1: Each submission creates a new lead (no dedup). Noted for v1.5 enhancement.
EC4	Brevo API is down during form submission	Lead is saved to Sheets (no data loss). Email flags remain FALSE. Agent can see the lead on dashboard.
EC5	Agent enters wrong PIN 10+ times	v1: No lockout mechanism. Each attempt simply returns "Invalid PIN." Note for v2: add rate limiting or lockout after N attempts.
EC6	Agent checks/unchecks "Contacted" rapidly	Each toggle sends an API call. Last state wins. No race condition — sequential processing in Apps Script.
EC7	Google Sheets is at capacity (10M cells)	Extremely unlikely for this use case (~10K leads = 110K cells). If it happens, Apps Script returns error.
EC8	Dashboard loaded with 0 leads	Stats show zeros, charts show empty state with "No data yet" message, table shows "No leads found" placeholder.
EC9	Brevo daily limit exceeded (300 emails/day)	Brevo returns 429 error. Lead is saved, email not sent. Agent is notified via dashboard error message.
EC10	Customer enters extremely long text in Area field	Client-side: max 100 chars enforced. Server-side: truncated to 100 chars before saving.

8. Summary of All Email Types

#	EMAIL NAME	RECIPIENT	TRIGGER	FREQUENCY	TEMPLATE SUBJECT
EM1	Lead Acknowledgment	Customer	Form submission (automatic)	Once per lead	"Thank you for your interest, {name}!"
EM2	New Lead Notification	Agent	Form submission (automatic)	Once per lead	"🏠 New Lead: {name} — ₹{price} in {area}"
EM3	Uncontacted Reminder	Agent	Scheduler (hourly check)	Once per lead (flag-gated)	"🕒 Reminder: {name} hasn't been contacted"
EM4	Agent Follow-up	Customer	Agent clicks "Send Email"	Manual (can be multiple)	"Regarding your property inquiry — {agentName}"